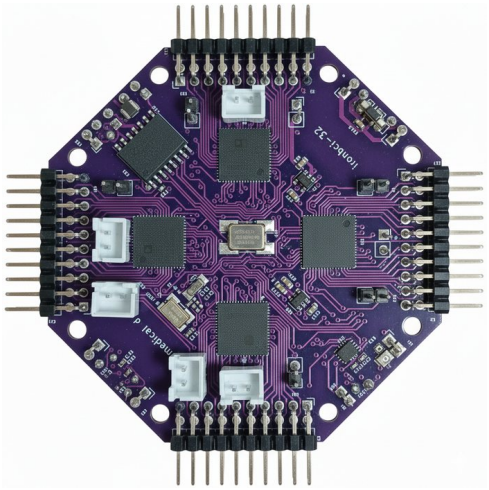
PRODUCT DATASHEET · SKU PIA24025P

32-Channel EEG Acquisition Board

Low-cost, open-source 32-channel, 24-bit biosignal acquisition board with four Analog Devices AD7771 converters and an STM32H7 microcontroller. Thirty-two simultaneous channels of EEG, EMG, ECG or EOG stream to a laptop over a single USB cable.



32	24-bit	~500 SPS	USB	70 × 70
Channels + REF + BIAS	4 × AD7771 converters	Per channel, all 32	CDC · 921 600 baud	mm · 4-layer PCB

01 OVERVIEW

Key features

- **Full-head coverage** — 32 synchronised channels on the 10-20 system, plus reference and bias.
- **Quad AD7771 front end** — four 8-channel, 24-bit sigma-delta converters sampling all channels at the same instant.
- **Ultra-low noise** — low-noise sources and references; noise measured below 0.22 µV.
- **STM32H743 (Cortex-M7)** — packs every sample into one frame with real-time pre-processing.
- **Plug and stream** — starts streaming on power-up; no start command, no driver on Windows 10/11, macOS or Linux.
- **Open ecosystem** — native BrainFlow board (IRONBCI_32_BOARD), PiEEG Server, PiEEG Cloud; open KiCad, Gerber and firmware.

Typical applications

- High-density EEG and ERP research
- Brain-computer interfaces and motor decoding
- Source localisation and topographic mapping
- Teaching labs and neuro-engineering courses
- EMG, ECG and EOG prototyping
- Neurofeedback and signal-processing development

02 TECHNICAL SPECIFICATIONS

ANALOG ACQUISITION	
Channels	32 biosignal inputs in 4 groups of 8, plus reference and bias
Analog front end	4 × Analog Devices AD7771, 8 channels each, simultaneous sampling
Resolution	24-bit
Input noise	Ultra-low-noise sources and references; measured below 0.22 µV
Sample rate	About 500 SPS per channel, fixed by firmware (BrainFlow board description: 512 Hz)
Reference / bias	REF and BIAS ear-clip inputs on the CH 1–8 header
µV scaling	2.5 V reference, gain 8: 1 LSB ≈ 0.03725 µV
Signals	EEG, EMG, ECG, EOG
DIGITAL & INTERFACE	
Microcontroller	STMicroelectronics STM32H743, Arm Cortex-M7, real-time pre-processing

For research and education only. Not a medical device. Specifications subject to change.

Host link	Micro-USB “Data Stream”, USB CDC-ACM virtual COM port, 921 600 baud
Frame format	0xA0 · counter · 32 × 3 bytes · status · 0xC0 (107 bytes on current firmware)
Programming	SWD (GND, RST, SWDIO, SWCLK) for ST-Link; STM32CubeProgrammer
ELECTRODES	
Electrode type	Dry or wet Ag/AgCl
Connectors	4 × 2.54 mm double-row (2 × 10) right-angle headers, one per AD7771
Recommended cap	PiEEG 32-channel wet cap (matching channel layout)
POWER & SAFETY	
Supply	5 V from a USB battery (power bank) only, on the dedicated Micro-USB “Power” port
Isolation	None. The board and the host laptop must both run on batteries while electrodes are attached
Classification	Research and education tool. Not a medical device
MECHANICAL	
PCB	70 × 70 mm regular octagon, 4 layers, 1.6 mm FR-4
Mounting	6 × Ø3.2 mm non-plated holes (M3)
Enclosure	Open board; 3D-printable cases on Thingiverse (PiEEG/designs)
SOFTWARE	
BrainFlow	BoardIds.IRONBCI_32_BOARD (ID 65) — Python, C++, Java, C#, Julia, R, MATLAB
PiEEG	PiEEG Server (pip install "pieeg-server[ironbci32]"), PiEEG Cloud (Web Serial), JavaScript SDK
Open files	KiCad schematic, Gerber files, firmware .hex and STM32H743 source (MIT-licensed SDK)

03 BLOCK DIAGRAM

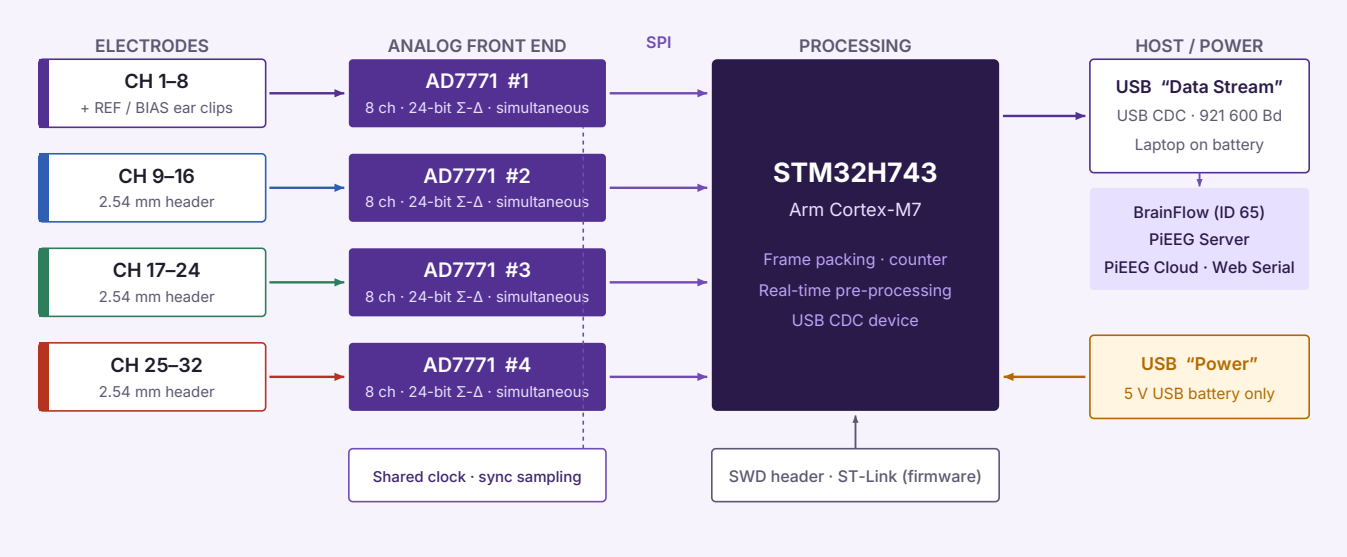


Figure 1 — Signal path. Each AD7771 digitises one header (8 channels); the STM32H743 combines all four into one frame per sample and streams it over USB.

04 MEASURABLE SIGNALS

Signal	What it shows	Typical first test
EEG	Brain rhythms across the whole scalp	Close the eyes and watch alpha (8–13 Hz) rise over the occipital channels
EMG	Muscle activity	Clench the jaw or a fist
ECG	Heart activity	Electrodes on the chest or wrists
EOG	Eye movement and blinks	Look left and right, blink

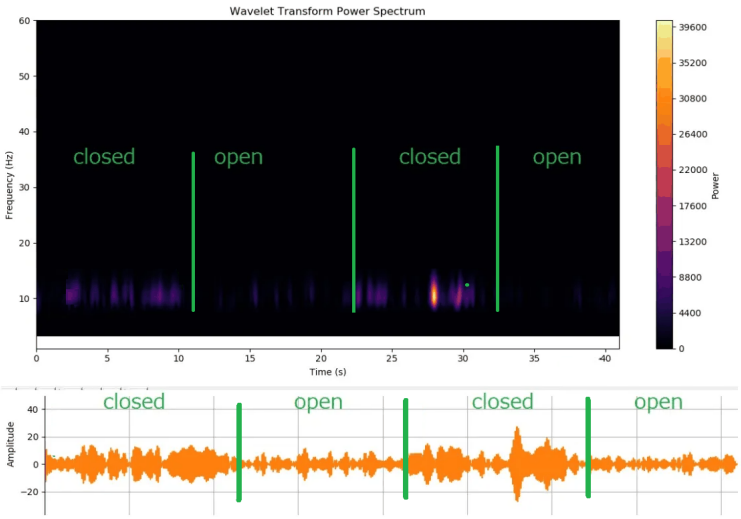


Figure 2 — Alpha test with dry Ag/AgCl electrodes (no gel): eyes closed / open with wavelet power spectrum.

Example recording

The repository ships a 32-channel, 20-second recording made with the PiEEG 32-channel wet cap (`dataset/eeg_32ch_20s.xlsx`, 10 241 samples per channel, 512 Hz). See the Data Interface & Software Guide (IRB32-SW-006) for traces and spectra.

Raw values carry a large DC offset from the electrode contact (tens of thousands of µV). Apply a high-pass or 1–40 Hz band-pass filter before reading EEG.

05 COMPARISON WITH OTHER PIEEG BOARDS

	PiEEG-16	IronBCI-8	IronBCI-32
Channels	16	8	**32**
Host	Raspberry Pi	None (Bluetooth)	Laptop or desktop (USB)
Converters	2 × ADS1299	1 × ADS1299	4 × AD7771
Sample rate	250 SPS	250 SPS	**~500 SPS**
BrainFlow	No	No	**Yes**
Best for	Pi-based research	Wearable, mobile	Full-head, high-density EEG

06 ORDERING INFORMATION

Product	IronBCI-32 — 32-channel professional research system
SKU	PIA24025P
Purchase	Elecrow (official manufacturing partner) — elecrow.com/ironbci-32.html
Institutions	5+ units: 10–20 % discount, pro-forma invoices and PO support via pieeg.com

Included	Details
1 × IronBCI-32 board	4 × AD7771, STM32H743, 2 × Micro-USB, 4 × 2.54 mm headers
Reference / bias interface	2.54 mm interface for 32-channel caps
Software	BrainFlow support, open-source firmware and example scripts

Not included — required	
5 V USB battery	Powers the board
2 × Micro-USB cables	One must be a full data cable
Laptop on battery	Windows, macOS or Linux
Electrodes / cap	32 Ag/AgCl + 2 ear clips, gel

WARNING Battery only, for everything.

IronBCI-32 has no isolation circuitry. Power the board from a 5 V USB battery and run the laptop on its own battery, unplugged from the wall. Never use a wall charger, a desktop PC or a mains-powered USB hub while electrodes are on a person. IronBCI-32 is not a medical device; do not use it for diagnosis or treatment.

07 RELATED DOCUMENTS

Doc. No.	Document	Contents
IRB32-MD-002	Mechanical & Dimensional Drawing	Outline, mounting holes, connector positions
IRB32-PCB-003	PCB Technical Information	Stack-up, layers, drill table, fabrication data
IRB32-PIN-004	Connector Pinout & Electrode Guide	Headers, pin order, REF/BIAS, 10-20 layout
IRB32-QSG-005	Quick Start & Safety Guide	Setup, battery-only operation, troubleshooting
IRB32-SW-006	Data Interface & Software Guide	Serial frame, µV conversion, SDKs, firmware